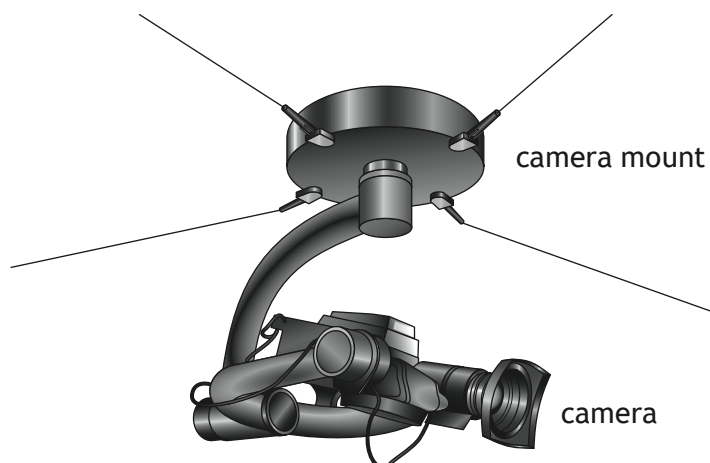
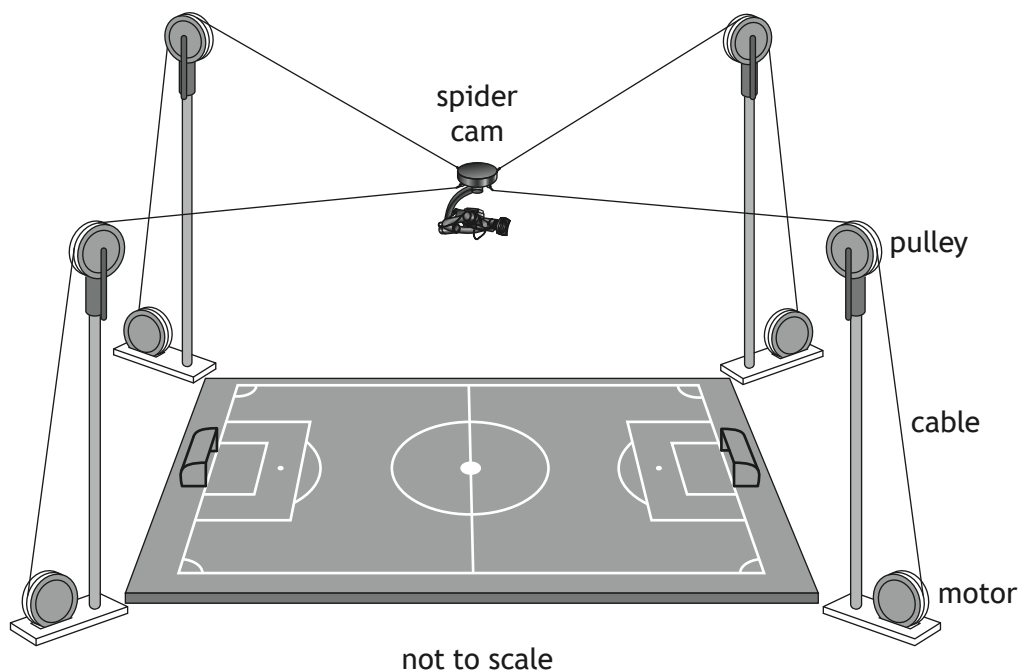


Total marks — 130
Attempt ALL questions

1. A television company uses a 'spider cam' to film a sporting event. A 'spider cam' is made up of a camera mount that carries a camera.



The camera mount is suspended from four cables. The length of the cables can be changed using a motor attached to each cable. This allows the height and position of the camera over the pitch to be changed.



1. (continued)

The camera mount and camera have a maximum acceleration of 1.45 m s^{-2} and a maximum velocity of 8.86 m s^{-1} .

- (a) (i) State what is meant by an *acceleration of 1.45 m s^{-2}* .

1

- (ii) Calculate the distance travelled by the camera mount and camera as they accelerate at 1.45 m s^{-2} from rest to the maximum velocity.

3

Space for working and answer

[Turn over



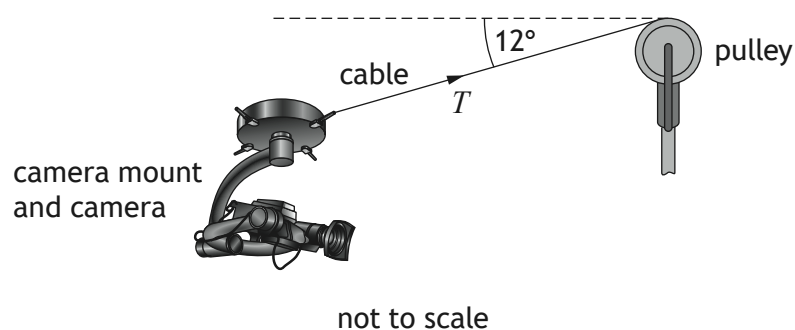
* X 8 5 7 7 6 0 1 0 5 *

1. (continued)

- (b) During a test of the system, the camera mount and camera are held stationary at a constant maximum height above the centre of the pitch.

The total mass of the camera mount and camera is 31 kg.

At the maximum height, the angle that **each** cable makes to the horizontal is 12° .



Determine the tension T in each of the four cables.

Space for working and answer

4